

Homework (Habanero)

Q1. What is $2^3 \cdot 2^4$?

- (A) 2^{3+4}
- (B) 2^{3-4}
- (C) $2^{3 \cdot 4}$
- (D) $2^{3/4}$
- (E) $2^{4/3}$

Q2. Simplify $(3^2)^3$ to get?

- (A) 3^{2+3}
- (B) 3^{2-3}
- (C) $3^{2 \cdot 3}$
- (D) $3^{2/3}$
- (E) $3^{3/2}$

Q3. If a medicine's dosage is 2^5 mg, and the doctor wants to increase it by a factor of 2^2 , what is the new dosage?

- (A) 2^{5+2} mg
- (B) 2^{5-2} mg
- (C) $2^{5 \cdot 2}$ mg
- (D) $2^{5/2}$ mg
- (E) $2^{2/5}$ mg

Q4. A cell grows at a rate of 3^2 per hour. If it starts with 3^2 cells, how many will it have after 2 hours?

- (A) 3^{2+2}
- (B) 3^{2-2}
- (C) $3^{2 \cdot 2}$
- (D) $3^{2/2}$
- (E) $3^{2 \cdot 2} \cdot 3^2$

Q5. What is the value of $\frac{2^6}{2^2}$?

- (A) 2^{6+2}
- (B) 2^{6-2}
- (C) $2^{6 \cdot 2}$
- (D) $2^{6/2}$
- (E) $2^{2/6}$

Q6. If $x = 2^3$, what is x^2 ?

- (A) 2^{3+2}
- (B) 2^{3-2}
- (C) $2^{3 \cdot 2}$
- (D) $2^{3/2}$
- (E) $2^{2/3}$

Q7. Simplify $\frac{(2^2)^3}{2^4}$ to get?

- (A) 2^{2+3-4}
- (B) 2^{2-3-4}
- (C) $2^{2 \cdot 3-4}$
- (D) $2^{2/3-4}$
- (E) $2^{2/3}$

Q8. A medical trial has 2^5 participants. If 2^3 more join, what fraction of the original number have joined?

- (A) $\frac{2^3}{2^5}$
- (B) $\frac{2^5}{2^3}$
- (C) $\frac{2^{3+5}}{2^5}$
- (D) $\frac{2^{5-3}}{2^3}$
- (E) $\frac{2^{5+3}}{2^5}$

Open Ended Questions (FRQ)

Q9. A new medicine's effectiveness is modeled by $y = 2^x$, where y is the percentage of patients cured and x is the dosage. If $x = 3$, what percentage of patients will be cured? Show your work.

Q10. The growth rate of a bacterium is given by $y = 3^{0.5x}$, where y is the number of bacteria and x is time in hours. If after 2 hours there are 9 bacteria, what was the initial number of bacteria? Show your steps.

Chef's Tips

- Tip 1: Check units and significant figures.
- Tip 2: Label all diagrams and graphs clearly.
- Tip 3: Use the given formulae to derive your answer.